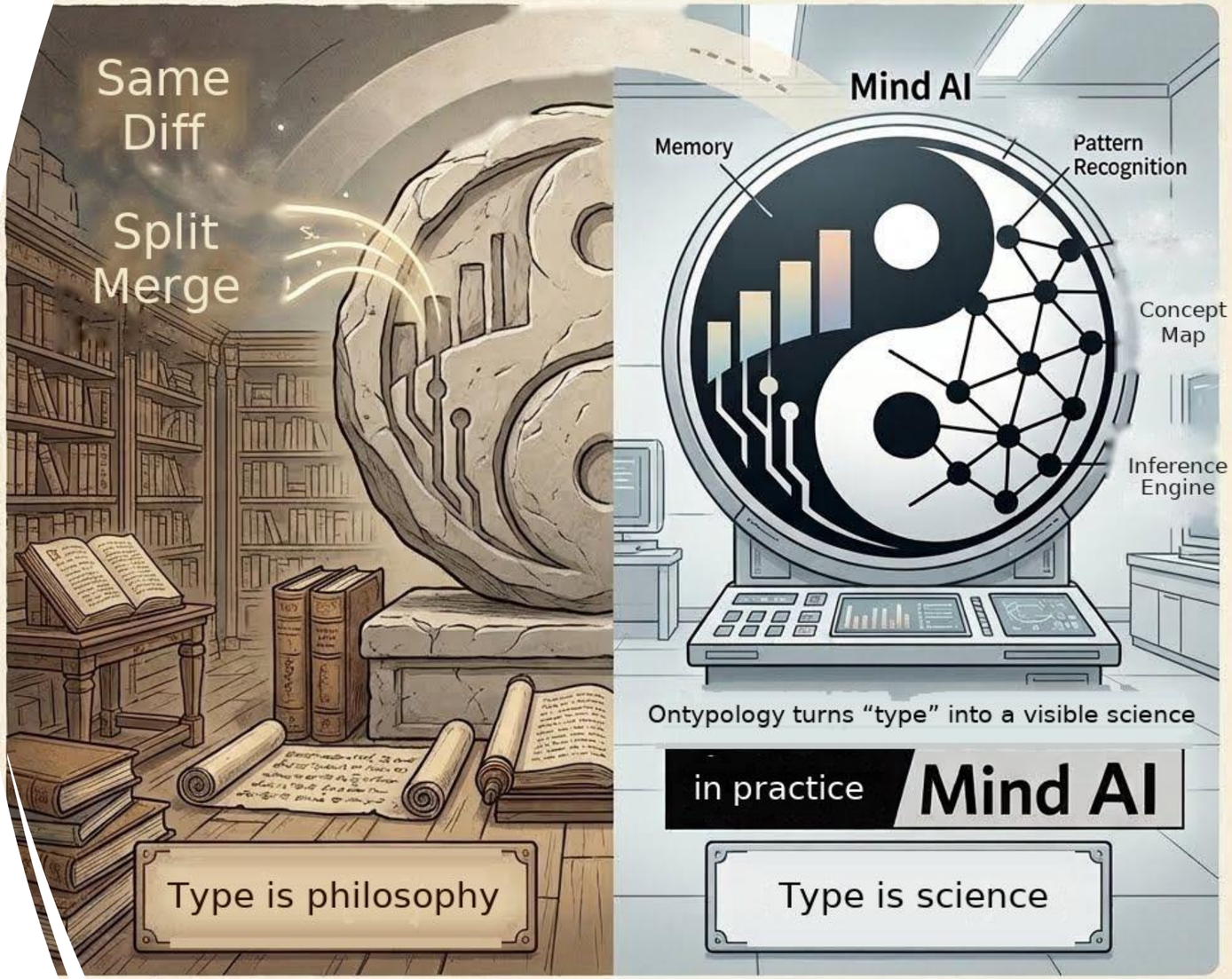
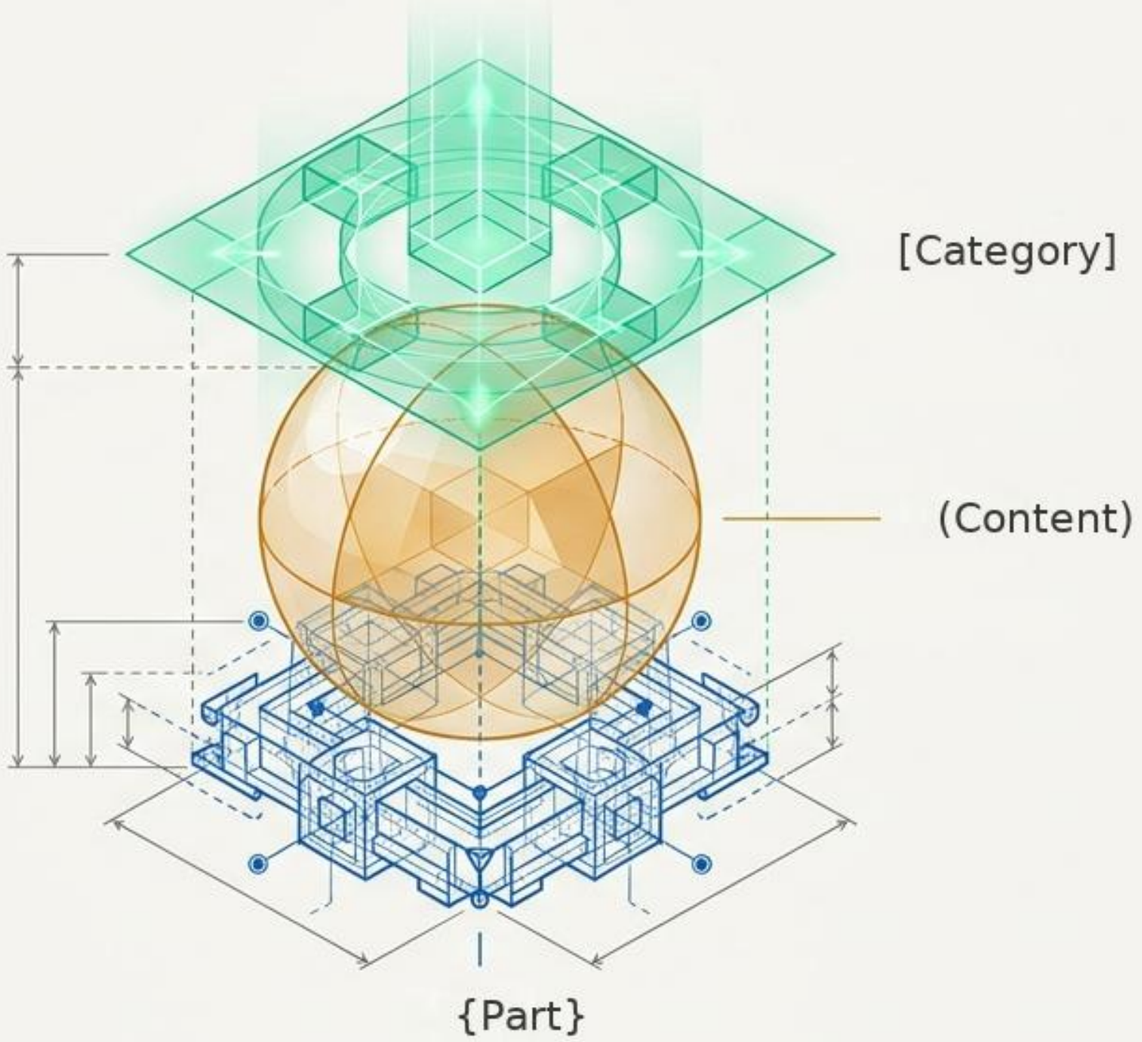
Ontypology

- From sameness and difference to splitting and merging, from splitting and merging to type, from type to understanding — the brain walked this road for billions of years.
- Logic was philosophy a century ago; once seen clearly it became a leading discipline (science)
- Today the inquiry into "type" (a brain concept) is philosophy; Ontypology makes "type" visible and turns it into a leading discipline (science), and puts it into practice with Mind AI.

Ontypology ★

Same/Diff → Split/Merge → Type → Understanding → Science



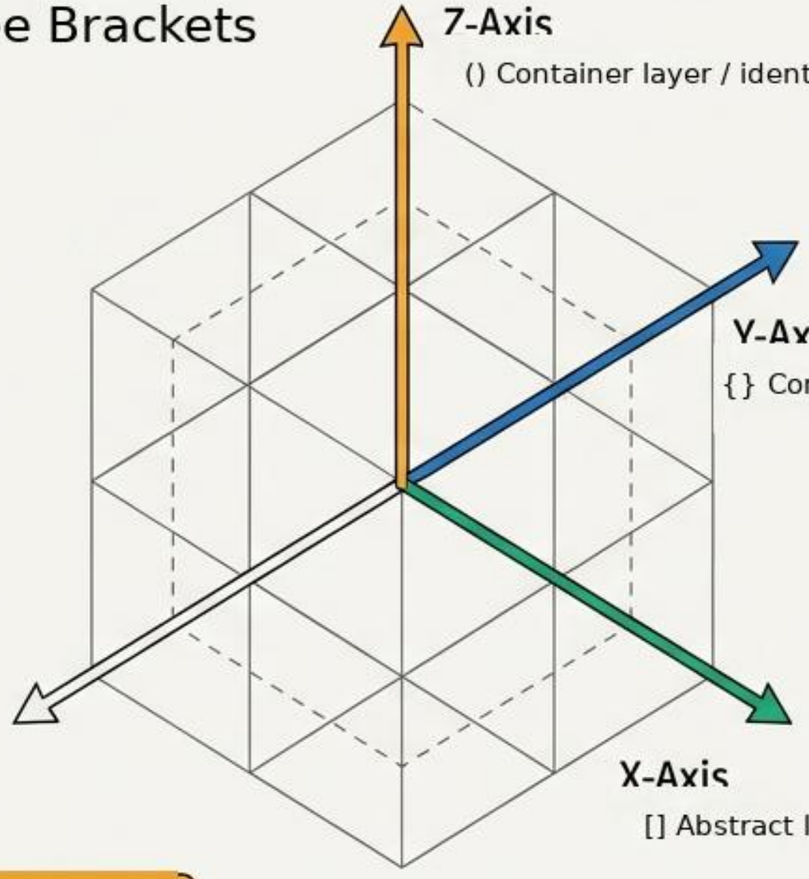


The Expression of Type Sets

From "Static Notation" to "Dynamic Derivation"
— building a checkable blueprint

AI Master-Level Visual Guide

Three Axes, Three Brackets

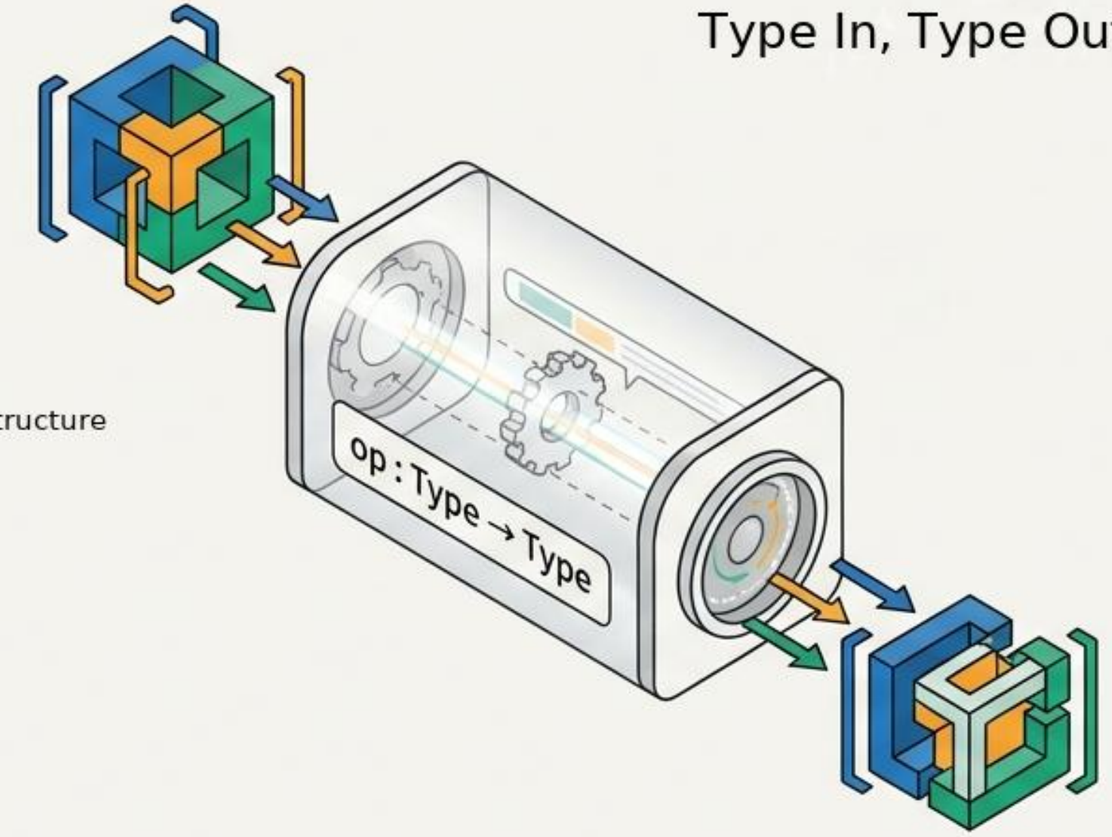


Operands

Type Set M = {Part, (Content), [Category]}.

Brackets fix the operands; operands fix the operations.

Type In, Type Out



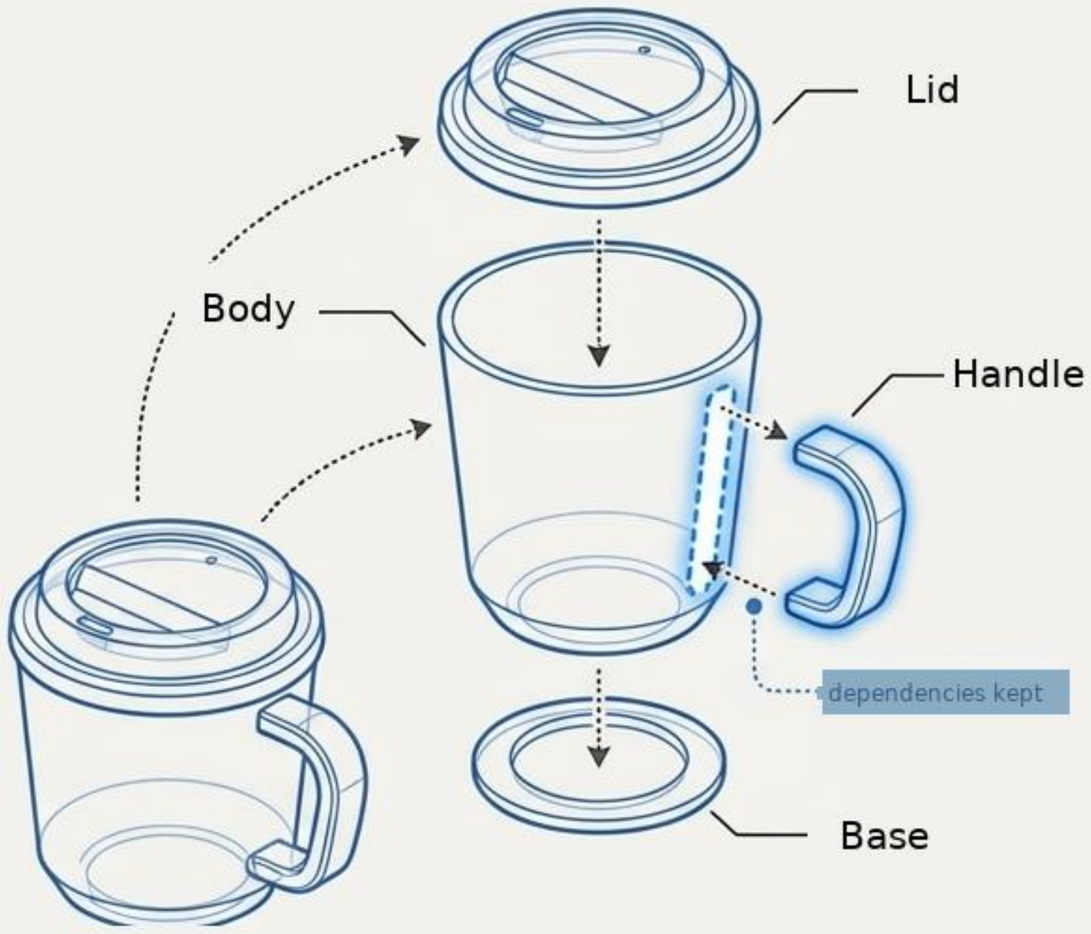
Two Cornerstones

1. Tuple criterion: T = (F(T), N(T)) and C = (id(C), S(C))
2. Axiom of invariance: replace content, keep identity.

Core principle: every operation is type-in / type-out — valid only inside its brackets.

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Tool I: Combine



Definition

Merge the parts into a single type.

All parts absorbed; hierarchy and dependency fully kept.

Law Check

[Holds] ✓ Associativity: (A⊕B)⊕C ≡ A⊕(B⊕C)

(the assembly order does not change the final type)

[Fails] ✗ Commutativity: A⊕B ≠ B⊕A

Counter-example

"Reorganizing departments": Sales Dept. vs. Admin Dept.

Swap the order, rewire the relations — that is another organizational type.

Position and dependency are part of the type.

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Tool II: Remove

Core rule: removal is not the inverse of combination.

Dependencies built by combination leave only a fracture behind.

Three-tier degradation map



M (smartphone)

remove(p, M)

Remove mainboard (essential)



≠ M

(disassembled)

Remove camera (degradable)



≡ M′ ⊂ M

(function degraded)

Remove protector (non-essential)



≡ M

(no effect)

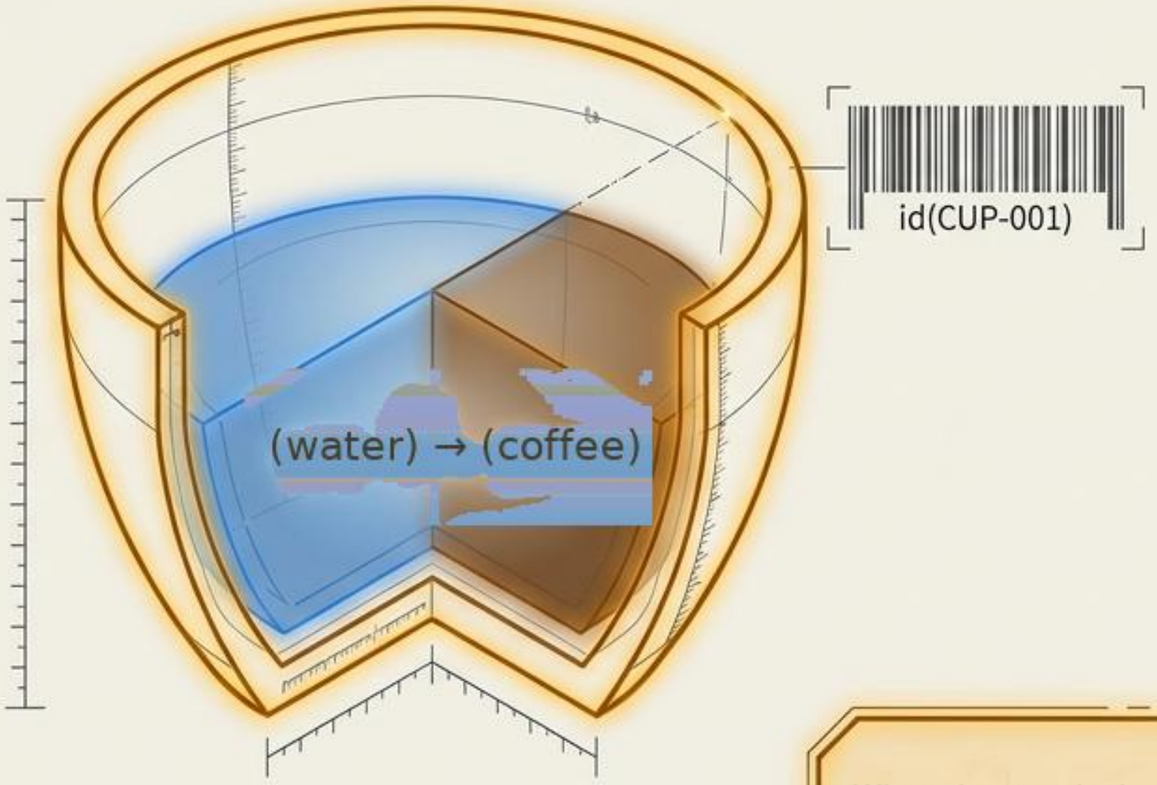
Idempotence: remove(p, remove(p, M)) = remove(p, M)

(a second removal finds no part — the result is unchanged).

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Tool III: Substitute

$$\sigma(e1 \to e2)(C)$$



Before / After	
✓	Correct: swap water for coffee. id unchanged (identity-preservation theorem).
✗	Cross-level misuse: swap the cup for a bowl. This touches the id — it is disassembly, not a σ operation. Brackets decide the level!

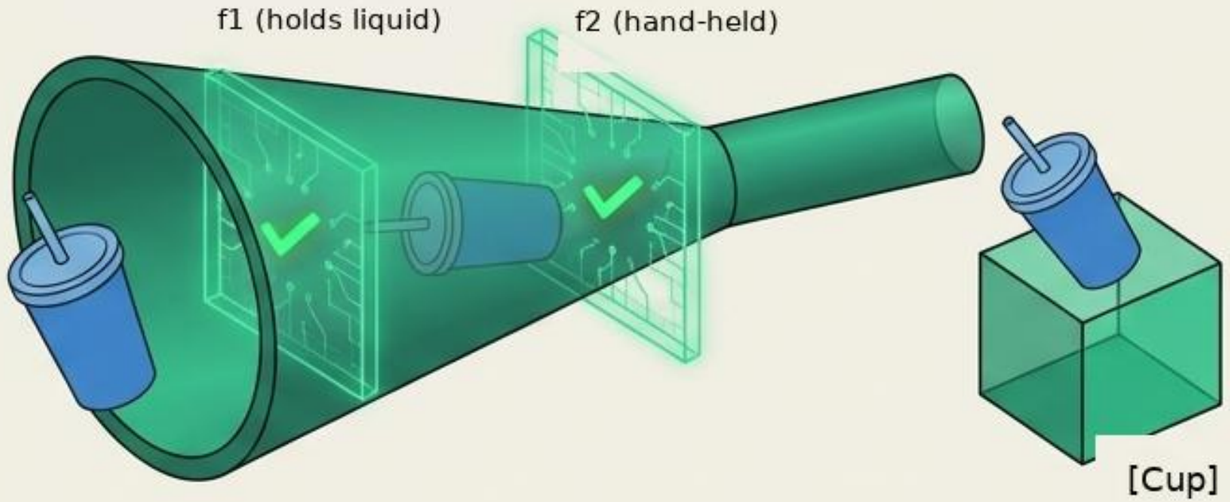
What changes is the content; what stays is the identity. (Axiom of invariance)

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Tool IV: Classification ∩(T) and Abstraction α

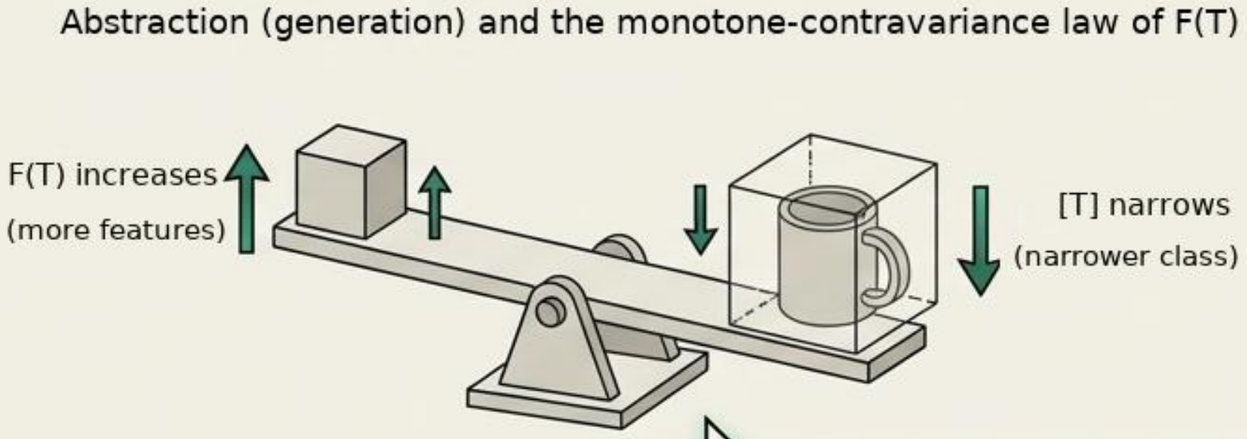
Isometric Schematics & Optical Filters

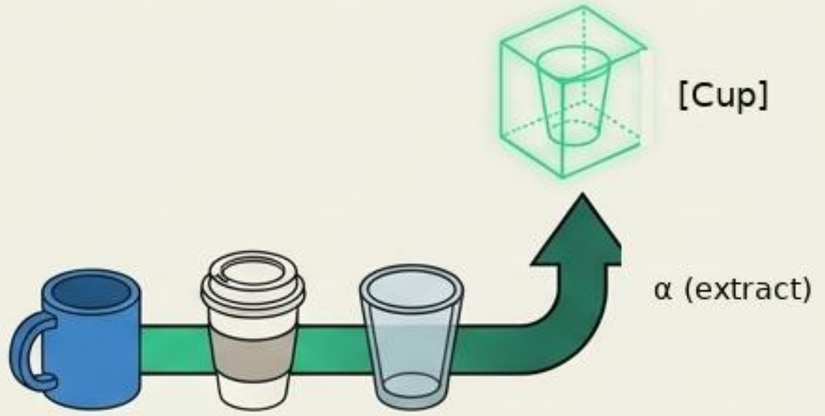
Classification (judgment)



Only when every feature passes (∧),
can it enter the class. Order is irrelevant.

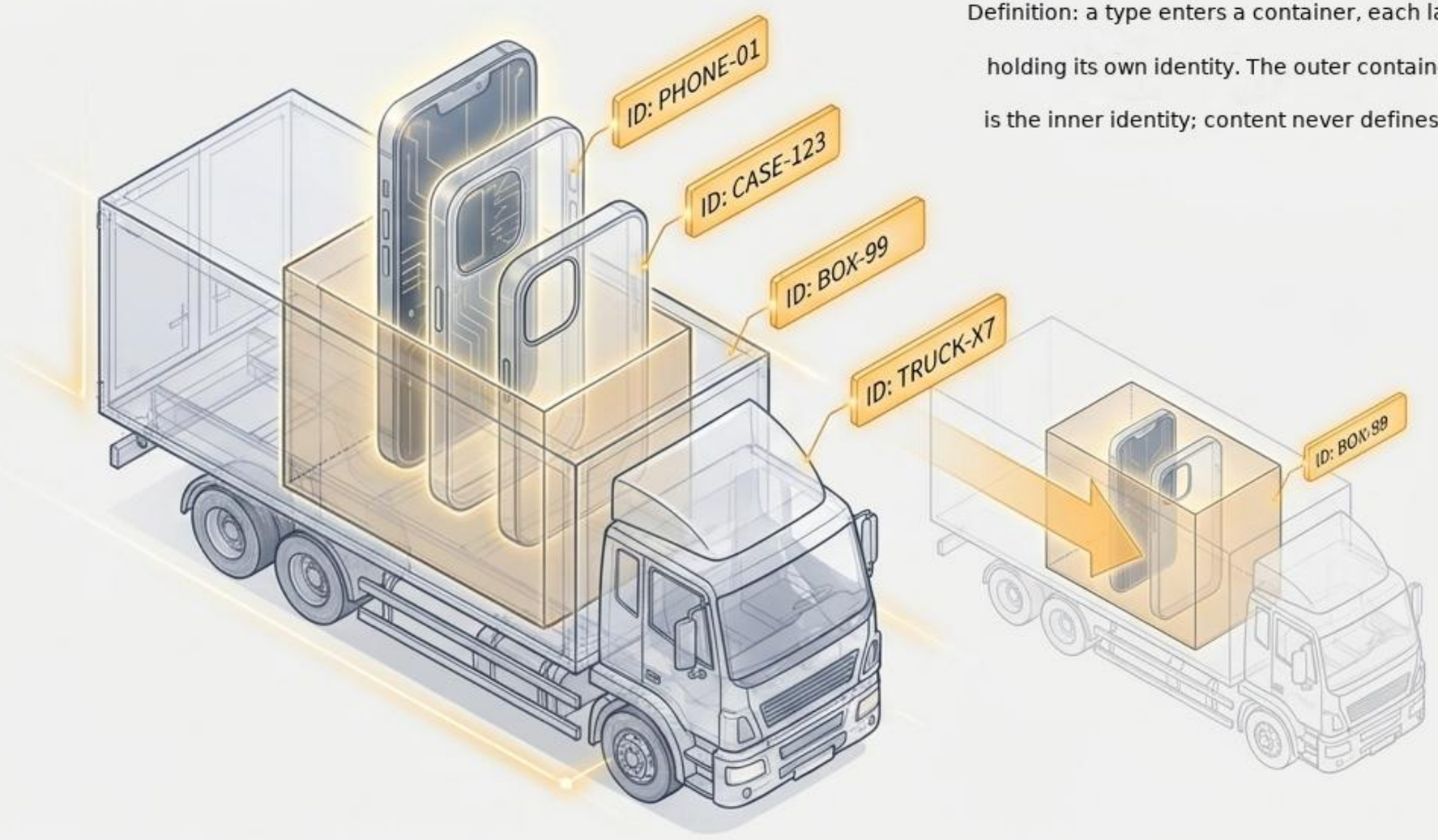
Abstraction (generation) and the monotone-contravariance law of F(T)





Features add up, the class narrows — monotone contravariance.

Tool V: Nesting τ (Nest)



Definition: a type enters a container, each layer holding its own identity. The outer container is the inner identity; content never defines it.

Pitfall Zone

No-Flattening Rule

✗ {Cup, water}

flattened into a flat list. identity scatters and loses its meaning.

✅ (CUP-001, (water))

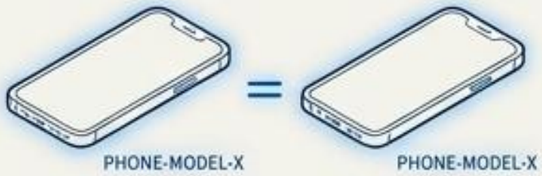
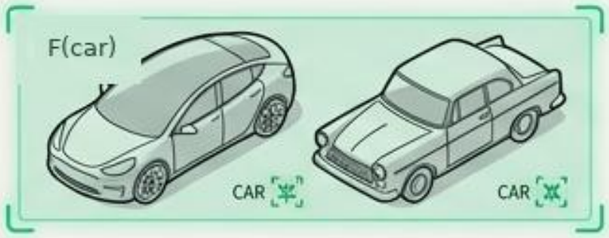
Keep both the container and content layers; only then can it work.

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Criterion: Comparison ≡ / ≠ / ⊂

Before comparing, ask: which whole am I comparing, and identity at which layer?

Three kinds of identity — a comparison matrix

Container layer (same id) Container Amber		water → coffee	≡ (same CUP-001) Container Amber
Composition layer (same type) Structural Blue		two phones of exactly the same model	✓ ≡ (BOM and structure fully match) Structural Blue
Abstract layer (same class) Abstract Emerald		Tesla vs. petrol car	✓ ≡ (both pass F(car)) Abstract Emerald



Degradation: C (M′ ⊂ M) — type identity survives,
but the function is a subset. Not ≠ (disassembly).

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Law Boundaries: The Ultimate Law Matrix

Behind every counter-example hides one constraint — "layer" and "identity". Break it and the formula fails.

✅ Safe zone (holds)



Associativity (layer-independent)



remove: idempotent

(removing twice changes nothing)



σ preserves identity (axiom of invariance)



F(T) monotone contravariance

! Conditional zone (holds with a condition)



and commutativity

↓

substitution must happen inside

Forbidden zone (never holds)



Commutativity

Out of bounds: position is part of the type



remove: essential part

Out of bounds: outright disassembly



Cross-layer swap

Out of bounds: new container ≠ new content

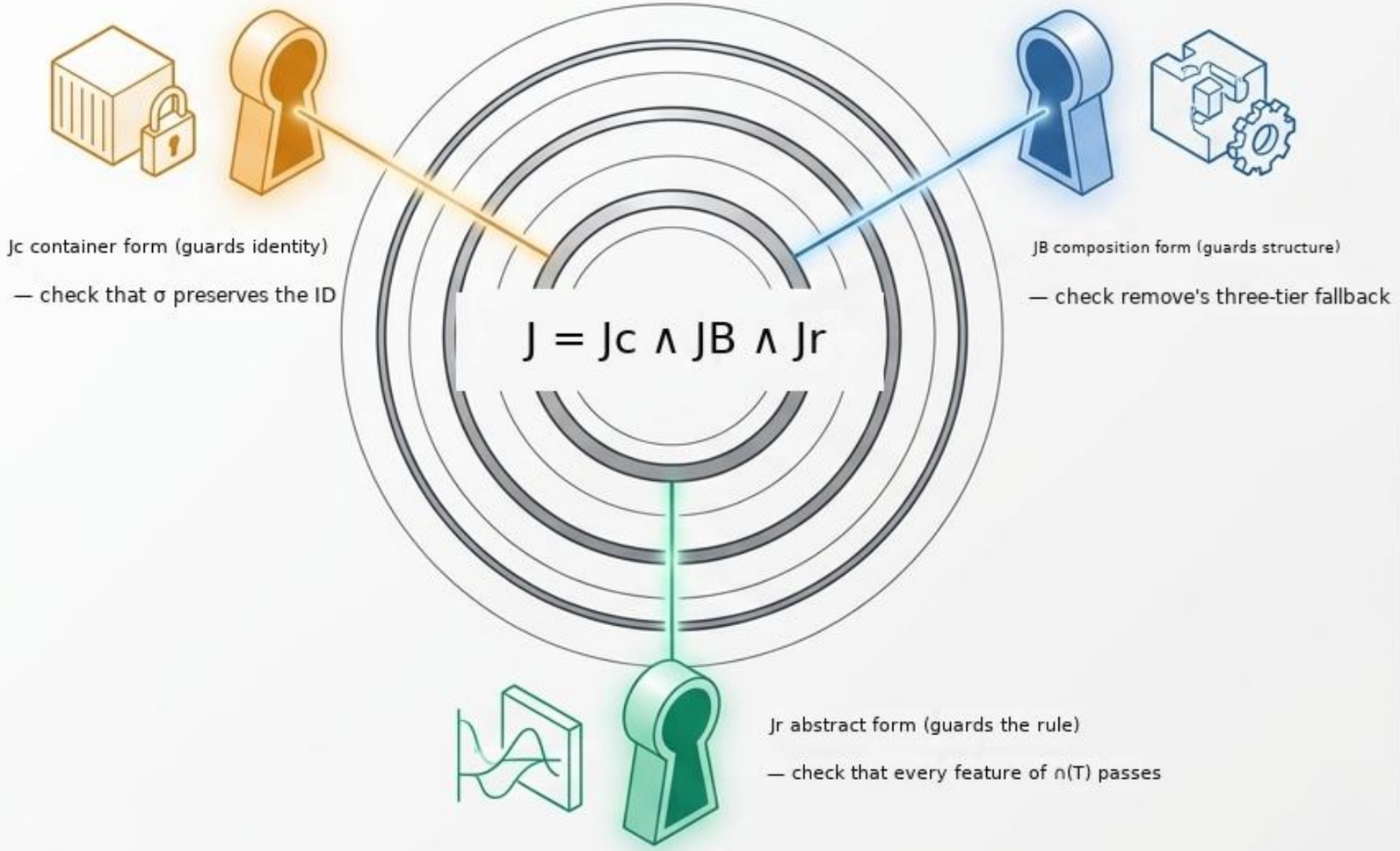


remove-then-abstract ≠ abstract-then-remove

sample variance

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Final Defence: Making the Three Formulas Computable



Jc container form (guards identity)

— check that σ preserves the ID

JB composition form (guards structure)

— check remove's three-tier fallback

Jr abstract form (guards the rule)

— check that every feature of ∩(T) passes

Worked example (smartphone)



Blind spot of Jc alone:

a tablet also runs apps.



Blind spot of JB alone:

a calculator shares a similar part structure.



Blind spot of Jr alone:

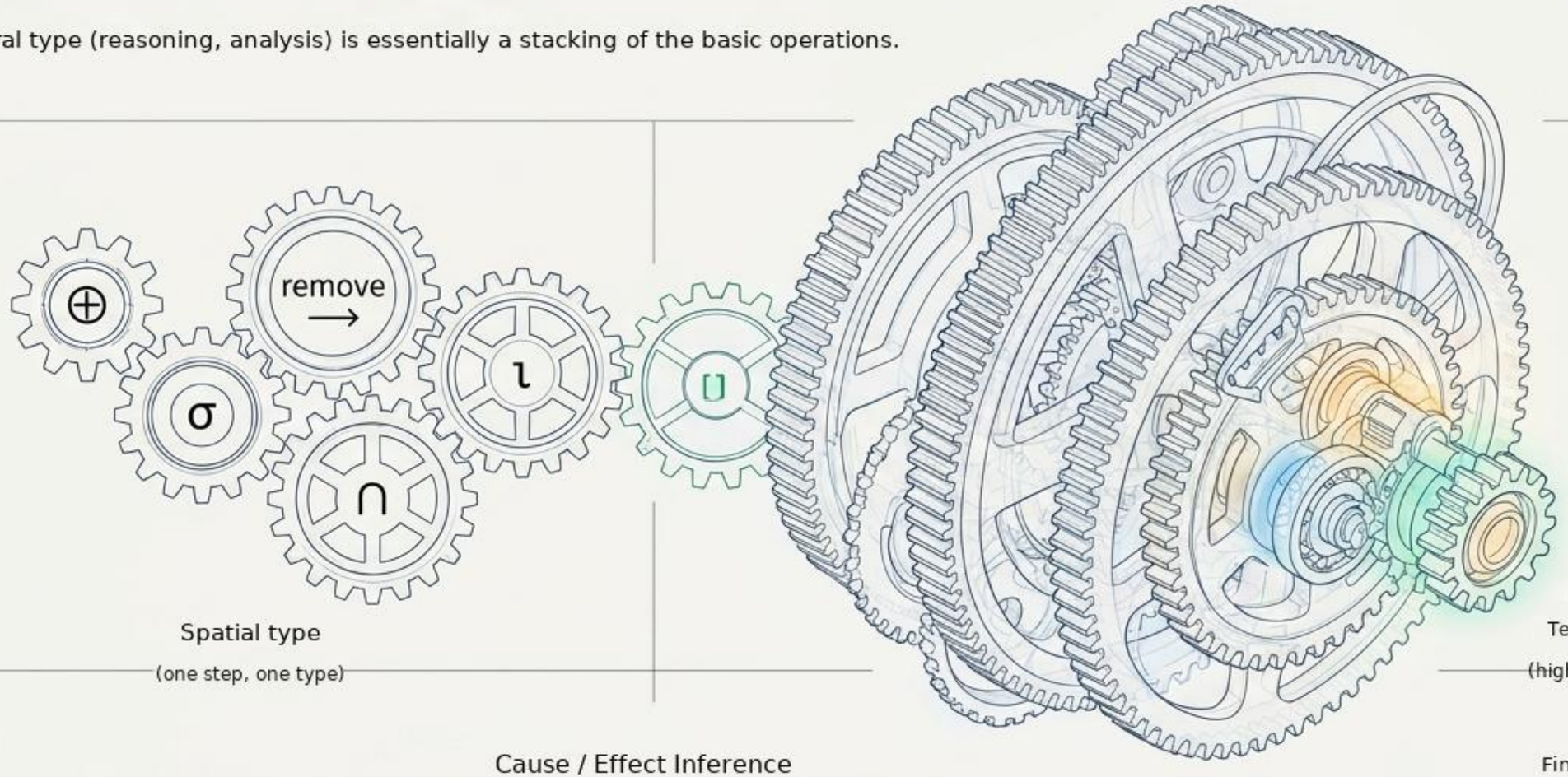
walkie-talkies also communicate.

The three formulas complement one another. Turn all three keys and only then does it light up as proven (True). An error passed by one layer is always caught by the other two.

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Higher-Order Composition: Reaching Temporal Types

Every temporal type (reasoning, analysis) is essentially a stacking of the basic operations.



Spatial type

(one step, one type)

Gear linkage

Temporal type library

(higher-order composition)

{ } Component layer

( ) Container layer

[ ] Category layer

Cause / Effect Inference

Finding the lost lid



Observe

remove (degraded)

Trace back

(reverse the nesting chain: desk > office)

State match σ

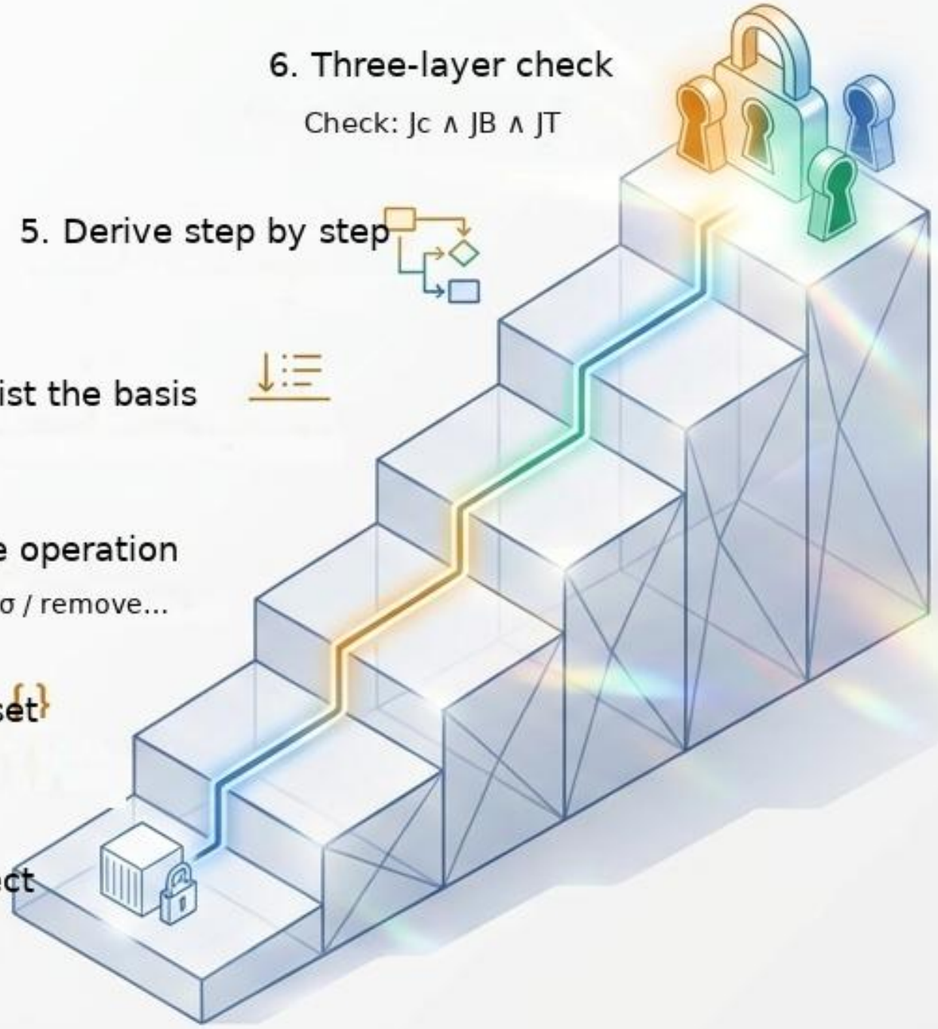
(check the substitution log)

=

(lock the ID to find the cup)

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Field Manual: The Six Derivation Steps



8-point self-check before delivery (Checklist)

- [ ] Is the object clear?
- [ ] Is the type set complete?
- [ ] Do object and operation match?
- [ ] Are bracket scopes marked? (no ambiguity)
- [ ] Do the cited laws hold?
- [ ] Have counter-examples been sought? (confirm the boundary)
- [ ] Can a reader tick each line and reproduce it?
- [ ] Has the three-layer check been done?

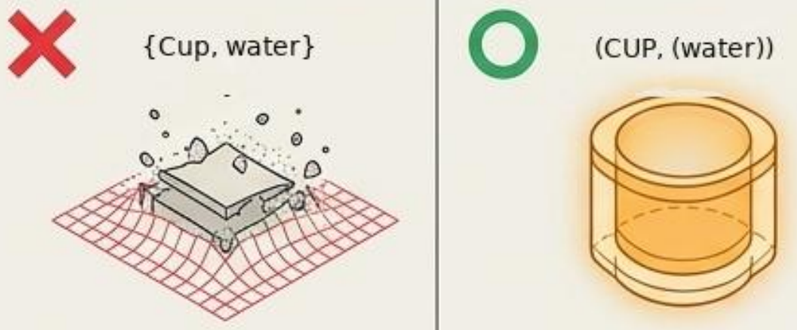
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Pitfall Guide: Five Fatal Misuses

Common root: the layer was never made explicit, so the brackets overreach.

Error-diagnosis board

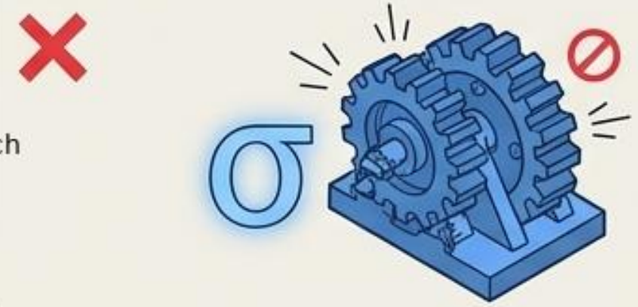
1. Layer flattened



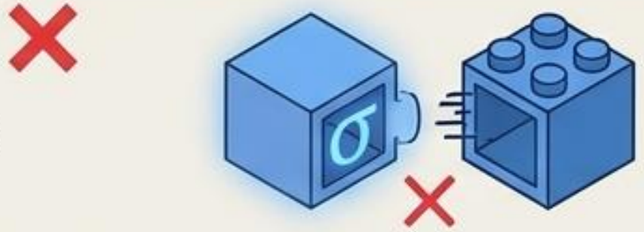
2. Wrong object



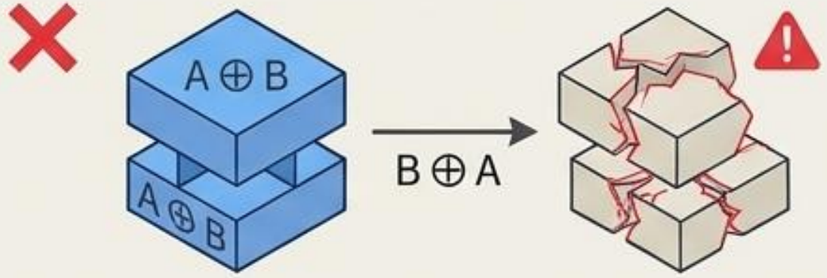
3. Operation mismatch



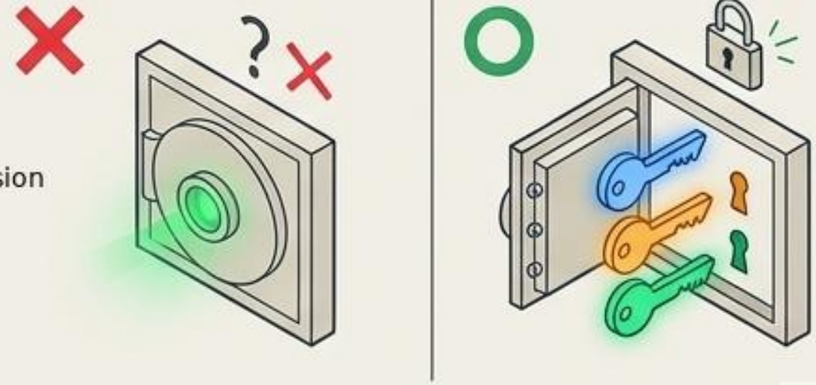
3. Operation mismatch



4. Law overreach

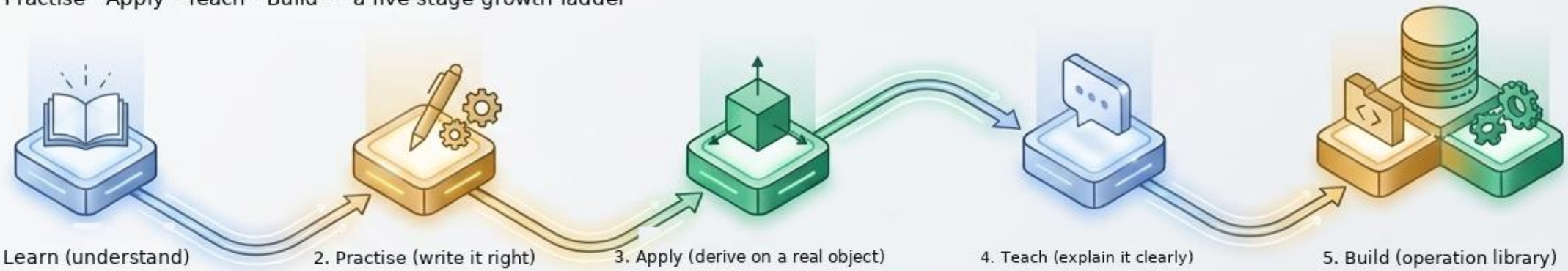


5. Single-layer conclusion

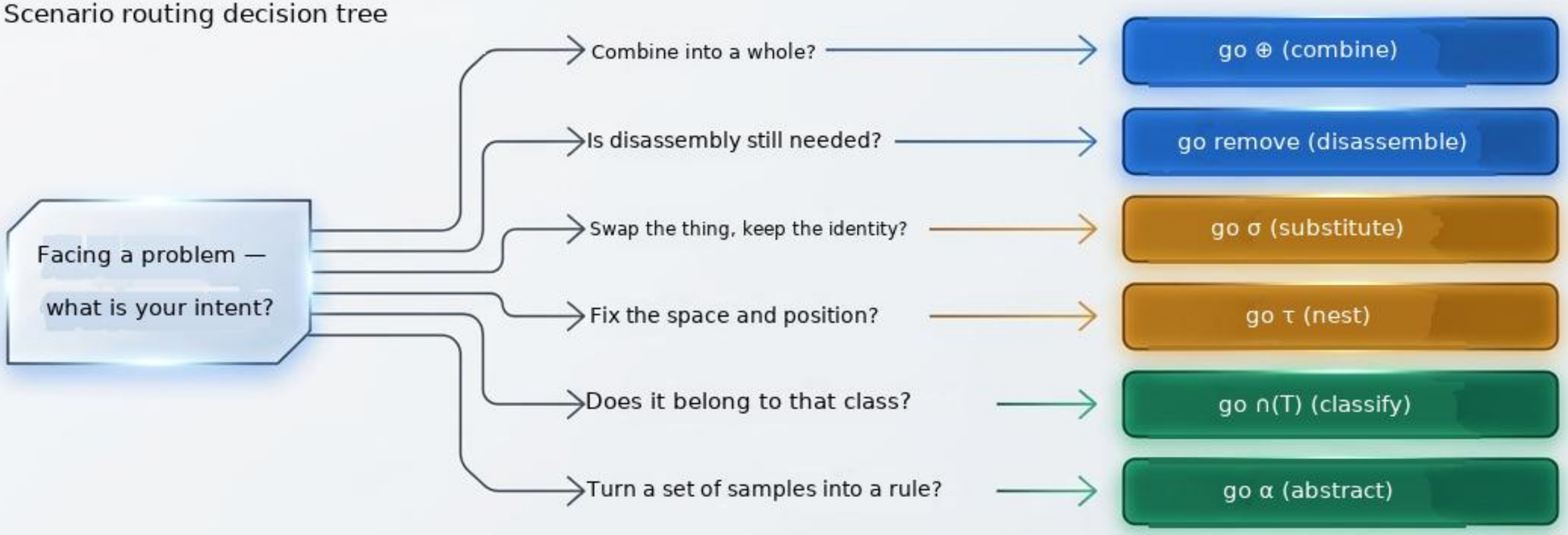


Putting It to Work: Scenario Routing and the Growth Ladder

Learn · Practise · Apply · Teach · Build — a five-stage growth ladder



Scenario routing decision tree



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Ultimate Cheat Sheet

Symbol table

 ⊕	Combine	{} part
 remove	Remove	{} part
 σ	Substitute	() content
 ι	Nest	() content
 ∩(T)	Classify	[] category
 α	Abstract	[] category
 ≡ / ⊂	Compare & degrade	all three layers

Terminology

type in, type out: op: Type → Type

Axiom of invariance: replace content, keep the container ID

Monotone contravariance: features add, the class narrows

No flattening: keep container and content layers

Derivation templates

Single-form derivation:

[object] -> [operation] -> [basis] -> [derivation] -> [conclusion]

Law check:

[formula] -> [verdict] -> [derivation or counter-example]

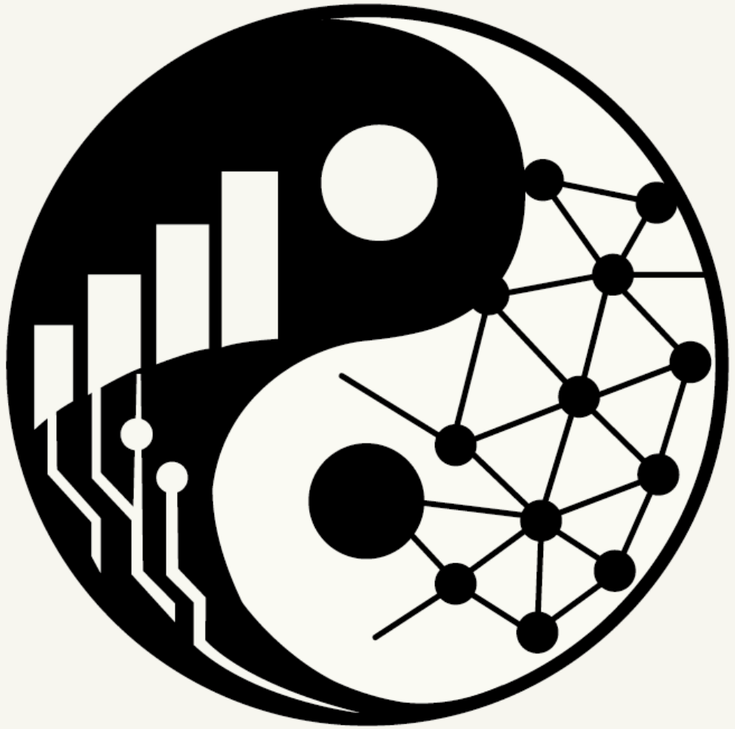
Three-form card:

[J_C] ∧ [J_B] ∧ [J_T] = proven

Notation is static; operation is dynamic — see the type clearly, make the formula exact.

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LonGreat vision: intelligent life that understands the world



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"Let AI be more than a tool — a living intelligence that can truly understand, reason, learn and create."

From unified knowledge to unified thinking, and finally to unified intelligence — this is the eternal path of AGI that LonGreat pursues.

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